

Independent Study

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This Week

- Improved neurosymbolic pipeline for accuracy and cost
- Recreating LTRAG's results

Setting up pipeline for testing

I modified LTRAG to adapt to reasoning model testing. I included a reasoning effort initializer in setup files and connected all parts of the codebase as necessary. I also modified my [runner.py](#) to match this new installment.

Problems during setup

I soon came to realize that after running a basic smoke test on a GPT-5 nano model, parts of my code didn't connect the way I wanted them to, and largely, the original LTRAG codebase wasn't very neat. Their standard and cot files were revised. I had to fix a few things, and I modified my [runner.py](#) to match these changes. I also made sure there would be a cost analysis of the tokens, which I extracted after every LLM call. I used OpenAI's website from the last deliverable for these calculations. After setting up everything and making sure the pipelines ran properly, I carried out larger tests.

Testing

I ran GPT-5.4 mini on reasoning = none and reasoning = medium for each of standard, COT, and LTRAG on the whole FOLIO dataset. This ran surprisingly quickly. The no reasoning took less than 3 minutes, while the medium took less than 10 minutes. This is by design because of the parallelism that exists in the original code. I was able to run 10 processes (10 samples) at once locally with no problems.

Results and Analysis

The results are under this [repo](#), then myLTRAG/5.4_mini.txt. The core part we should observe is none reasoning: COT accuracy is 62.1%, while LTRAG accuracy is 70.9%. However, LTRAG is significantly more expensive. What we want to compare LTRAG on is nothing to do with COT on medium, which achieved 70.9% accuracy as well. COT on medium is still more expensive than LTRAG on none, which we need to fix. However, we are close to attaining support for my initial hypothesis.

Conclusion

After running my first official test on my hypothesis, I am feeling hopeful because we are able to match COT with reasoning on non-reasoning. Next steps would be finding ways to make LTRAG more affordable and achieving better accuracy.

References

Token costs:

Openai website

AI Use:

I used Claude to help me with the following:

1. Help understanding OpenAI's API calling information.
2. Help with formatting and extracting JSON data for token costs
3. Help updating the format for printing in stdout
4. General debugging where appropriate

All changes were made by the author

